

Closed forms for the constants in the odd-weight identities between dihedral modular graph functions

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Abstract We study the combinations X_w of dihedral two-loop modular graph functions $C(a, b, c)$ of odd weight w whose Laplacian is proportional to the Eisenstein series E_w . We establish $X_w = f_w E_w + g_w \zeta(w)$ with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$ at the odd weights $3 \leq w \leq 61$. The argument combines exact rational Laplace algebra with the Laurent constant term evaluated in rational arithmetic from a published theorem. That theorem and the standard lemma on harmonic invariant functions of polynomial growth are assumed and not machine-checked. The harmonic space is exactly one-dimensional for odd w and zero for even w at the weights $3 \leq w \leq 25$. Independently of the proof, the identities at $w = 3, \dots, 17$ were confirmed numerically at 32 digits at verifier-chosen points; $w = 15$ was a prediction and $w = 17$ a preregistered blind test. The full weight-11 basis has exactly one relation. A hardening run exposed and fixed an evaluator bug. A proof beyond the weights examined remains open and is stated as a conjecture.

Keywords Modular graph functions · Laplace eigenvalue equations · Eisenstein series · Bernoulli numbers · Genus-one string amplitudes · Computer-assisted proof

1 Introduction

Dihedral modular graph functions $C(a, b, c)$ arise as coefficients in the low-energy expansion of the one-loop closed-string amplitude. D'Hoker, Green and Vanhove showed that they satisfy Laplace eigenvalue equations with inhomogeneous terms polynomial in non-holomorphic Eisenstein series, and they exhibited exact differential and algebraic relations for an infinite class of such functions [1]. The Laplace equation fixes an algebraic relation only up to an additive constant of integration, which is obtained from the asymptotics at the cusp [1]. In the simplest cases this gives $C(1, 1, 1) = E_3 + \zeta(3)$ and $C(2, 2, 1) = \frac{2}{5}E_5 + \zeta(5)/30$. At weight 7 the identity $C(3, 3, 1) + C(3, 2, 2) = \frac{3}{7}E_7 + \zeta(7)/252$ appears in a later hierarchy of modular graph identities [2]. Related relations between graphs with four and five links, and between higher-loop graphs, were also treated in the literature [3, 4].

Open question. For odd w there is a combination X_w of $C(a, b, c)$ (eq. 3.57 of arXiv:1502.06698) whose Laplacian is proportional to E_w . The constants f_w and g_w in $X_w = f_w E_w + g_w \zeta(w)$ were given in the

literature only for small weights, with g_7 and g_9 left as undetermined integration constants. Do the constants have closed forms, and can these be established?

Why it is hard. The constant of integration requires the Laurent polynomial of each $C(a, b, c)$, and both the number of graphs and the size of the coefficients grow with w . Numerical determination by integer-relation searches is limited by working precision relative to coefficient height. An exact route is available through the Laurent polynomial of the two-loop modular graph functions (Theorem 5.1 of arXiv:1902.04180). That theorem was not evaluated for these identities there.

Contribution. We evaluate it in rational arithmetic and obtain closed forms for f_w and g_w at the odd weights $3 \leq w \leq 61$ examined. We also give an exact uniqueness statement and independent numerical confirmations.

Summary of results.

- Theorem 1 shows $X_w = f_w E_w + g_w \zeta(w)$ with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$ at the odd weights $3 \leq w \leq 61$ (exact; proof in three steps with two stated assumptions).
- Theorem 2 shows that the space $\{X : \Delta X \in \mathbb{Q}E_w\}$ is one-dimensional for odd w and zero for even w at the weights $3 \leq w \leq 25$ (exact).
- Conjecture 1 states the closed forms beyond the examined range; they are established only for $3 \leq w \leq 61$.
- The numerical observations confirm the identities independently of the proof at $w = 3, \dots, 17$. The full weight-11 basis has exactly one relation, and a hardening run is reported.

2 Setting

Let $\tau = \tau_1 + i\tau_2$, $q = e^{2\pi i\tau}$ and $p = m\tau + n$. The lattice-sum normalisation is $(\tau_2/(\pi|m\tau + n|^2))^a$. Primed sums exclude vanishing lattice vectors:

Both definitions follow. The hyperbolic Laplacian is $\Delta = \tau_2^2(\partial_{\tau_1}^2 + \partial_{\tau_2}^2)$, and E_w satisfies $\Delta E_w = w(w-1)E_w$. Holomorphic Eisenstein series are normalised by $a_0 = 1$, $E_k = 1 - (2k/B_k) \sum \sigma_{k-1}(n)q^n$; they enter only in Appendix A. X_w denotes the combination of dihedral $C(a, b, c)$ of weight w of D'Hoker–Green–Vanhove (eq. 3.57 of arXiv:1502.06698). The constants are

$$f_w = \frac{3((w-1)/2)!}{w}, \quad g_w = \frac{6|B_{w-1}|}{((w-1)/2)!},$$

with B_n the Bernoulli numbers.

Numerical checks use these parameters: 4 verifier-chosen points (seed 4711, $|\tau_1| \leq 1/2$, $1 \leq \tau_2 \leq 2.5$), working precision 32 digits, a maximal accepted relative residual of $1e-24$ for relations without Laplacian (1e-14 with Laplacian), and a fourth-order finite-difference Laplacian with step $h = 1e-6$.

Statements are labelled by their status: *exact* (rational arithmetic), *numerical* (verifier points), and *conjecture*.

3 Exact identities for odd weight

Theorem 1 (Closed forms of the constants, $3 \leq w \leq 61$). *At every odd weight $3 \leq w \leq 61$, $X_w = f_w E_w + g_w \zeta(w)$ holds identically in τ , with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$. In particular $g_w = 1, 1/10, 1/42, 1/120, 1/264, 691/327600, 1/720, 3617/3427200$ at $w = 3, 5, 7, 9, 11, 13, 15, 17$ respectively.*

Proof. The proof has three steps. (i) Exact rational computation in the algebraic Laplace representation gives $\Delta X_w = w(w-1)f_w E_w$ with no other terms, at the odd weights $w \leq 61$. (ii) Since $\Delta E_w = w(w-1)E_w$, the function $X_w - f_w E_w$ is harmonic. It is $SL(2, \mathbb{Z})$ -invariant and of polynomial growth at the cusp, hence constant by the standard argument of D'Hoker–Green–Vanhove. (iii) The constant equals the τ_2^0 term of the Laurent polynomial of X_w , which Proposition 2.1 and Theorem 5.1 of D'Hoker–Kaidi (arXiv:1902.04180) give in closed form. Evaluated in rational arithmetic it equals $g_w \zeta(w)$. The transcription was checked against their eq. (5.19). In every term not involving products of zeta values, the Laurent polynomial of X_w agrees with that of $f_w E_w + g_w \zeta(w)$, which is a consistency check. *Assumptions not checked by code:* correctness of Theorem 5.1 of arXiv:1902.04180, which is proved there in an appendix and whose conjectural part is not used, and the standard lemma in step (ii).

Status: partly known. The combinations X_w for odd w (DGV eq. 3.57) are known, as are f_w for $w \leq 9$ and g_w for $w = 3$ and $w = 5$ (eq. 3.34, with g_7 and g_9 undetermined there), and g_7 is also explicit in eq. 3.77 of arXiv:2109.05017. Not given there are closed forms of f_w, g_w at odd w and their proof for $w \geq 9$.

The constants at the lower weights are tabulated below.

Here g_w is the coefficient of $\zeta(w)$.

Proposition 1 (Constants for $27 \leq w \leq 35$). *At the odd weights $3 \leq w \leq 35$ one has $\Delta X_w = w(w-1)f_w E_w$ and the Laurent constant $g_w = 6|B_{w-1}|/((w-1)/2)!$.*

Certificate. Exact Laplace algebra and rational evaluation of the Laurent polynomial of Theorem 5.1 of arXiv:1902.04180, with the stated values equal to $6|B_{w-1}|/((w-1)/2)!$. This is the intermediate certificate run before Theorem 1. *Status:* partly known, with the same literature status as Theorem 1.

Remark 1 (Status of the computer-assisted parts). Trusted base: the rational (exact fraction) arithmetic

w	f_w	g_w	claim
3	1	1	
5	6/5	1/10	
7	18/7	1/42	
9	8	1/120	
11	360/11	1/264	
13	2160/13	691/327600	
15	1008	1/720	
17	120960/17	3617/3427200	
19	1088640/19	43867/48263040	
21	518400	174611/199584000	
23	119750400/23	77683/83462400	
25	57480192	236364091/217945728000	

of the checker; the Laplace representation of D'Hoker–Green–Vanhove, re-derived and checked in rational arithmetic; and, as mathematical inputs not verified by code, Theorem 5.1 of arXiv:1902.04180 and the standard lemma of step (ii) of Theorem 1. The numerical statements below use 32-digit arithmetic with an exact leading-coefficient test in $y = \pi \tau_2$ plus residuals at verifier-chosen points. A third party re-runs every certificate with the commands listed under Code and data availability and in Appendix A.

4 Uniqueness and Laplace algebra

Theorem 2 (Uniqueness of the harmonic combination, $3 \leq w \leq 25$). *At every weight $3 \leq w \leq 25$, the rational combinations X of dihedral $C(a, b, c)$ of weight w with $\Delta X \in \mathbb{Q} E_w$ form a space of dimension 1 for odd w and 0 for even w . For odd $w = 2\mu + 3$ it is spanned by the DGV combination X_w , and $\Delta X_w = w(w-1)f_w E_w$ with $f_w = 3((w-1)/2)!/w$.*

Proof. Exact rational linear algebra in the algebraic Laplace representation, carried out at the weights $3 \leq w \leq 25$. *Status:* partly known. The one-dimensionality of the eigenspace at odd weight and its vanishing at even weight, together with X_w , are in the literature [1]. The closed form $f_w = 3((w-1)/2)!/w$ is not given there, and the exact confirmation is an independent recomputation.

Proposition 2 (Laplace equations at $w = 9$ and $w = 11$). $\Delta(9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3)) = 288E_9$, and $\Delta(2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3)) = 100E_{11}$. Hence $9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3) - 4E_9$ and $2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3) - \frac{10}{11}E_{11}$ are annihilated by Δ .

Known as well is X_w for odd w (eq. 3.57), with f_w, g_w not given there.

Remark 2 (Conjecture 1: beyond the examined range). For odd $w \geq 3$, $X_w = f_w E_w + g_w \zeta(w)$ with $f_w = 3((w-1)/2)!/w$ and $g_w = 6|B_{w-1}|/((w-1)/2)!$. We expect that a proof beyond the examined range reduces to a binomial-sum identity for the Laurent constant term; this is not carried out here.

5 Independent numerical confirmation

The following statements are numerical. They do not use Theorem 1, and each passes the exact leading-coefficient

test in $y = \pi\tau_2$ and residuals at 4 verifier points.

Example 1 (Weights 3 and 5). $C(1, 1, 1) - E_3 - \zeta(3) = 0$ and $30C(2, 2, 1) - 12E_5 - \zeta(5) = 0$, as functions of τ . Maximal relative residuals are 5.13×10^{-44} and 1.57×10^{-43} .

Certificate. The exact leading coefficient in y^3 resp. y^5 vanishes; numerical residuals at 4 verifier points, seed 4711, 32 digits. *Status:* already known (DGV eq. 1.7).

Numerical observation 1 (Weight 7). $252C(3, 3, 1) + 252C(3, 2, 2) - 108E_7 - \zeta(7) = 0$, i.e. $C(3, 3, 1) + C(3, 2, 2) = \frac{3}{7}E_7 + \zeta(7)/252$, with maximal relative residual 1.59×10^{-43} .

Certificate. The exact leading coefficient in y^7 vanishes; 4 points, 32 digits. *Status:* already known (eq. 3.77 of arXiv:2109.05017), i.e. f_7 and g_7 .

Numerical observation 2 (Weight 9). $960E_9 + \zeta(9) - 2160C(4, 4, 1) - 4320C(4, 3, 2) - 960C(3, 3, 3) = 0$, with maximal relative residual 8.33×10^{-44} .

Certificate. The exact leading coefficient in y^9 vanishes; 4 points, 32 digits. *Status:* partly known. The identity $9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3) = 4E_9 + g_9$ is in DGV eq. 3.34, but the value of g_9 is not given there.

Numerical observation 3 (Weights 11, 13, 15, 17). Each of the following holds as a function of τ .

- $w = 11$: $19008C(5, 5, 1) + 38016C(5, 4, 2) + 19008C(5, 3, 3) + 28512C(4, 4, 3) - 8640E_{11} - \zeta(11) = 0$.
- $w = 13$: $117936000C(6, 6, 1) + 235872000C(6, 5, 2) + 235872000C(6, 4, 3) + 188697600C(5, 5, 3) + 212284800C(5, 4, 4) - 54432000E_{13} - 691\zeta(13) = 0$.
- $w = 15$: $1555200C(7, 7, 1) + 3110400C(7, 6, 2) + 3110400C(7, 5, 3) + 1555200C(7, 4, 4) + 2592000C(6, 6, 3) + 6220800C(6, 5, 4) + 1244160C(5, 5, 5) - 725760E_{15} - \zeta(15) = 0$.
- $w = 17$: $51819264000C(8, 8, 1) + 103638528000C(8, 7, 2) + 103638528000C(8, 6, 3) + 103638528000C(8, 5, 4) + 88833024000C(7, 7, 3) + 222082560000C(7, 6, 4) + 118444032000C(7, 5, 5) + 148055040000C(6, 6, 5) - 24385536000E_{17} - 3617\zeta(17) = 0$.

Certificate. The exact leading coefficients in y^{11} , y^{13} , y^{15} , y^{17} vanish; 4 verifier points, 32 digits, with the residuals listed in the table below. Weight 15 was a prediction made before its value was read; weight 17 was a preregistered blind test. *Status:* partly known. Existence and combination of the identity for odd w (DGV eq. 3.57) are known, while f_w, g_w for $w \geq 11$ are not given there.

Here $\zeta(w)$ denotes the Riemann zeta value at w .

Figure 1 (to be generated from the table of f_w and g_w above): $\log |g_w|$ and $\log f_w$ against odd w , to show the growth of the constants; the plotted data are exactly the tabulated rational values.

Table 1 Maximal relative residuals of the numerical relations (4 verifier points, seed 4711, 32 digits; residual bound 1e-24).

w	$\zeta(w)$ coefficient in the relation	residual	claim
3	-1	5.13×10^{-44}	
5	-1	1.57×10^{-43}	
7	-1	1.59×10^{-43}	
9	+1	8.33×10^{-44}	
11	-1	1.21×10^{-43}	
13	-691	5.67×10^{-44}	
15	-1	6.46×10^{-44}	
17	-3617	5.49×10^{-44}	

Numerical observation 4 (Full weight-11 basis). The rational linear relations among $C(9, 1, 1)$, $C(8, 2, 1)$, $C(7, 3, 1)$, $C(7, 2, 2)$, $C(6, 4, 1)$, $C(6, 3, 2)$, $C(5, 5, 1)$, $C(5, 4, 2)$, $C(5, 3, 3)$, $C(4, 4, 3)$, E_{11} , $\zeta(11)$ form a space of dimension exactly 1, spanned by the weight-11 relation above.

On the normalised 20×12 matrix one singular value lies below 1e-24 (the largest small one is 1.11×10^{-43}) and 11 lie above 1e-16 (the smallest large one is 3.69×10^{-9}). *Status:* partly known, as for the weight-11 relation.

Numerical observation 5 (Criterion 1 failed, reported openly). For the full weight-11 basis, the originally preregistered criterion 1 (15 points, seed 4712, other singular values required above 1e-8) failed. The singular values were 5.175 , 2.785 , 0.04862 , 0.003131 , 0.0001604 , 6.701×10^{-5} , 9.579×10^{-6} , 2.823×10^{-6} , 1.981×10^{-7} , 5.754×10^{-8} , 4.149×10^{-9} , 7.591×10^{-44} .

Certificate. Singular values of the normalised value matrix as stated. Criterion 2, preregistered afterwards on new data, passed.

Numerical observation 6 (Hardening of the weight-11 identity (numerical)). An independent double-precision direct lattice summation agrees to at most 1e-12 relative at 8 of 8 points.

Certificate. Preregistered H10, 8 points planned and 8 completed; the expectation “residual decreases with precision” was operationalised as “residual at most 10^{-d} at every working precision d ”, an exact 0 being read as below rounding resolution.

The columns d_1, d_2, d_3 are the residuals at the working precisions of 30, 45 and 60 digits; the last column is the independent double-precision lattice sum.

The run exposed an evaluator bug: the Fourier sum for E_s stopped when a term vanished, which happens at $\tau_1 = 1/4$ where $\cos(2\pi N\tau_1) = 0$ for odd N . Before the fix that point showed a precision-independent residual of 3.5×10^{-10} . All points listed were computed with the corrected code, and no earlier test point was affected.

6 Discussion

The identities $X_w = f_w E_w + g_w \zeta(w)$ are fixed by two independent facts: a Laplace equation with the single source E_w , and a constant determined at the cusp. The

τ	d_1	d_2	d_3	lattice sum	claim
(0.500000, 0.866026)	2.15×10^{-42}	7.01×10^{-57}	1.3×10^{-71}	4.372×10^{-14}	
(-0.250000, 0.968246)	0.0	1.27×10^{-56}	1.13×10^{-71}	3.396×10^{-14}	
(-0.462905, 1.431467)	9.45×10^{-42}	6.71×10^{-57}	1.49×10^{-72}	1.615×10^{-13}	
(0.064009, 2.095482)	4.0×10^{-42}	6.5×10^{-57}	7.21×10^{-73}	6.527×10^{-14}	
(0.106134, 1.977393)	1.95×10^{-42}	1.08×10^{-56}	8.19×10^{-72}	6.972×10^{-14}	
(0.212603, 2.774679)	7.06×10^{-42}	4.89×10^{-57}	1.58×10^{-71}	5.108×10^{-14}	
(0.115505, 2.030193)	7.28×10^{-42}	0.0	1.74×10^{-71}	6.941×10^{-14}	
(0.258526, 2.959735)	3.09×10^{-42}	2.44×10^{-56}	6.19×10^{-72}	4.772×10^{-14}	

factorial in f_w comes from the exact Laplace algebra (Theorem 2). The Bernoulli number in g_w comes from the Laurent constant term (Theorem 1). The numerical confirmations at $w = 3, \dots, 17$ test the result independently, because they use only the relations themselves and not the Laurent polynomial of Theorem 5.1.

Limitations. The result at the weights examined rests on Theorem 5.1 of arXiv:1902.04180 and on the standard lemma of step (ii), neither of which is checked by code. The numerical statements concern the verifier points only.

What did not work. The originally preregistered criterion 1 for the full weight-11 basis failed, because one singular value lies below its threshold; it was reported openly and the afterwards-preregistered criterion 2 passed. The hardening run found a precision-independent residual at $\tau_1 = 1/4$, caused by an early-stopping bug in the Fourier sum for E_s ; it was fixed and all reported points use the corrected code. A direct integer-relation search over the full weight-11 basis found no relation and was aborted, presumably because the working precision was too low relative to the coefficient height. Further negative results are collected in Appendix A.

Open questions.

- (i) Is there a proof beyond the examined range, presumably reducing to a binomial-sum identity for the Laurent constant term (Conjecture 1)?
- (ii) What are the constants in the non-harmonic eigenvalue equations for the other combinations of $C(a, b, c)$, whose Laplace equations have product sources?
- (iii) Do analogous closed forms exist at higher loop order?

Declarations

Affiliation The author line supplied for this preprint carries no institutional affiliation.

Acknowledgements Computations, literature searches and parts of the manuscript were prepared with an automated verifier-gated laboratory built on

the AI system Claude (Anthropic); every statement was accepted only after verification by code; the authors are responsible for the content.

Code and data availability Repository <https://github.com/NinjaTurtlesHackathons/Hacknation>, branch `claude/modulformen-paper`, directory `projects/modular`. All certificates are reproduced with `python -m asd.recheck modular`, and the paper is rebuilt with `python -m asd.paper --domain modular`.

Competing interests None.

A Numerical methods and error control

Certificate types.

- **Exact Laplace algebra.** The action of Δ on $C(a, b, c)$ is applied in the algebraic representation of D'Hoker–Green–Vanhove in rational arithmetic; the result is $\Delta X = \lambda E_w$ with λ an exact rational.
- **Laurent constants.** The Laurent polynomial of X_w is computed in rational arithmetic from Proposition 2.1 and Theorem 5.1 of arXiv:1902.04180, transcription checked against eq. (5.19).
- **Numerical relation.** An exact leading-coefficient test in $y = \pi\tau_2$ is followed by residuals at 4 verifier points (seed 4711) at 32 digits, with a tolerance of $1e-24$ for relations and $1e-14$ with the finite-difference Laplacian of step $1e-6$.
- **Relation space.** Each relation is checked individually; completeness follows from a gap in the singular values of the normalised value matrix, evaluated at verifier points with seed 4712.

Trusted base. Exact rational arithmetic, the Laplace representation, and the two unchecked inputs named in Theorem 1. Numerical statements additionally depend on the working precision.

Verifier self-test (validation anchors). Classical and known identities were preregistered as system tests; all passed.

The first two identities concern theta functions as eta quotients, a classical subject [5]. The identity $1728\Delta(1) = E_4^3 - E_6^2$ expresses the cusp form Δ as a polynomial in E_4 and E_6 [6]. The identities for θ_{E8}

Table 2 Sturm-certified holomorphic identities.

identity	group	weight	Sturm bound	coefficients vanishing	claim
$\theta(1)^4 - \eta(2)^{20}\eta(1)^{-8}\eta(4)^{-8} = 0$	$\Gamma_0(4)$	2	1 (index 6)	q^0 to q^{11}	
$\theta(1)^2 - \eta(2)^{10}\eta(1)^{-4}\eta(4)^{-4} = 0$	$\Gamma_0(4)$	1	1/2 (index 6)	q^0 to q^{10}	
$\theta_{E8} - E_4 = 0$	$\Gamma_0(1)$	4	1/3 (index 1)	q^0 to q^{10}	
$\theta_{E8}^2 - \theta_{D16} = 0$	$\Gamma_0(1)$	8	2/3 (index 1)	q^0 to q^{10}	
$\theta_{E8}^2 - E_4^2 = 0$	$\Gamma_0(1)$	8	2/3 (index 1)	q^0 to q^{10}	
$\theta_{E8}^2 - E_8 = 0$	$\Gamma_0(1)$	8	2/3 (index 1)	q^0 to q^{10}	
$1728\Delta(1) - E_4^3 + E_6^2 = 0$	$\Gamma_0(1)$	12	1 (index 1)	q^0 to q^{11}	
$j\Delta(1) - E_4^3 = 0$	$\Gamma_0(1)$	12	1 (index 1)	q^0 to q^{11}	

and θ_{D16} and the identity $j\Delta = E_4^3$ are reproductions of classical results (no claim of novelty).

The eta-span table was not found in a targeted search of 33 abstracts on 2026-10-04.

The relation spaces are one-dimensional in each case.

Further numerical Laplace equations. At 4 verifier points, with a maximal relative residual below the Laplace tolerance of:

- $\Delta C(3, 1, 1) - 6C(3, 1, 1) - \frac{86}{5}E_5 + 4E_2E_3 - \frac{1}{10}\zeta(5) = 0$, with maximal residual 1.63×10^{-25} ;
- $\Delta C(2, 1, 1) - 2C(2, 1, 1) - 9E_4 + E_2E_2 = 0$, with 2.39×10^{-25} ;
- $\Delta C(2, 2, 1) - 8E_5 = 0$, with 6.99×10^{-26} ;
- $\Delta C(2, 2, 1) - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$, with 6.74×10^{-26} .

Such equations with sources polynomial in Eisenstein series are described in the literature [3, 1, 4].

Negative results. Four questions produced no verified claim on some attempts.

- **Weight-6 relations.** For rational relations among $C(4, 1, 1)$, $C(3, 2, 1)$, $C(2, 2, 2)$, E_6 , E_2E_4 , E_3^2 , E_2^3 , $\zeta(3)E_3$, $\zeta(3)^2$, no claim passed. The 12×9 value matrix has no singular value below $1e-24$, 8 above $1e-8$, and the smallest large one is 2.44×10^{-8} .
- **First eta-span attempts.** The first attempts at the eta-span table produced no verified claim; the table above was obtained later. The presumed cause is working precision that is too low relative to the coefficient height.
- **Weights 13 and 15, first attempts.** The first attempts on the constant formula produced no verified claim; the relations were later tested directly (see the observation above).

Counter-checks. In total 53 adversarial counter-checks were run: 4 passed (none contradicting the statement tested), 48 did not pass and 1 was not executable. The perturbations that were refuted include altered $\zeta(w)$ coefficients and altered Laplace eigenvalues.

Laboratory workflow. The laboratory ran 17 rounds with 37 verified statements and 4 negative results;

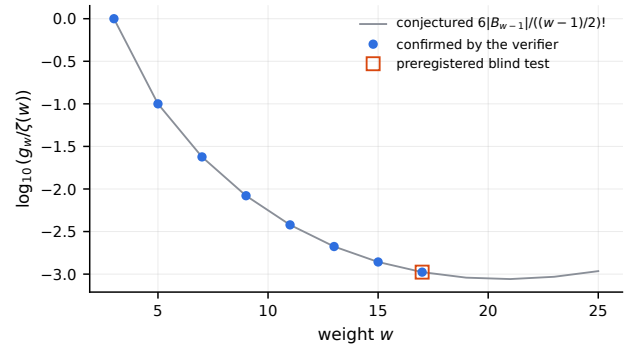


Fig. 1 Integration constants g_w of the odd-weight identities in the normalisation of eq. (3.57) of D’Hoker-Green-Vanhove: conjectured closed form (line) and the weights at which the verifier confirmed the identity numerically (dots, 32 digits, 4 verifier points); the square marks the preregistered blind test at $w = 17$.

every round was preregistered before the experiment (`prereg.md`). Statements were produced by agents in different roles (proposer, numerical checker, sceptic, theorist, a cheap baseline), and each was accepted only after verification by the code of the verifier. The sceptic role wrote the adversarial counter-checks counted above. Literature searches per result are recorded in the repository; for the blind test of the search used 92 abstracts.

B Provenance of the statements

The provenance table mapping every statement to its evidence follows (it is generated automatically).

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

References

- [1] Eric D’Hoker, Michael B. Green, and Pierre Vanhove. On the modular structure of the genus-one Type II superstring low energy expansion. *arXiv:1502.06698*, 2015.

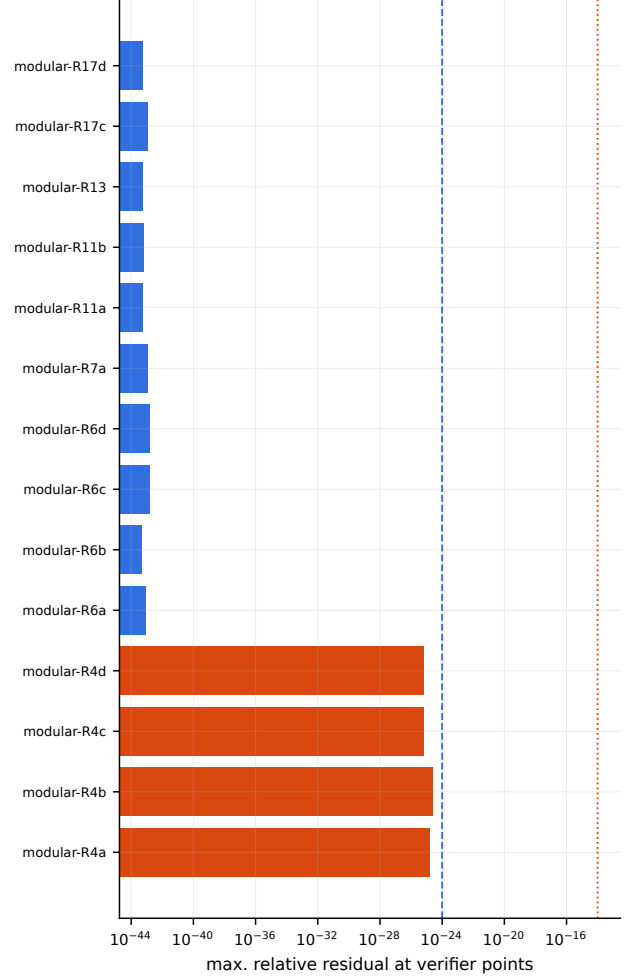
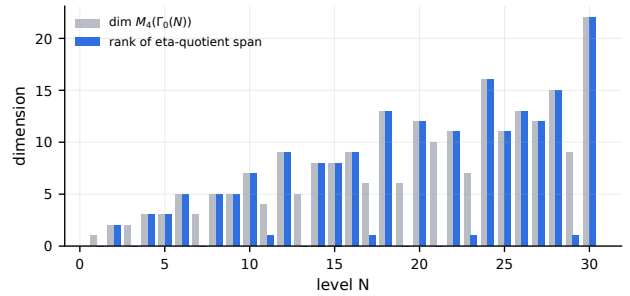
Table 3 Eta-quotient span of $M_4(\Gamma_0(N))$ for $N \leq 30$ (exact, exhaustive).

spans $M_4(\Gamma_0(N))$	does not span	claim
2, 4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, 24, 25, 26, 27, 28, 30	1, 3, 7, 11, 13, 17, 19, 21, 23, 29	

Table 4 Relation spaces at the weights 3, 5, 7, 9 (numerical; each relation checked individually with the exact leading Laurent coefficient; seed 4712).

w	matrix	null	large	smallest large	claim
3	6×3	1	2	0.731	
5	7×4	1	3	0.0372	
7	9×6	1	5	0.000127	
9	12×9	1	8	8.6×10^{-7}	

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**Fig. 2** Numerical verification margins: largest relative residual of each confirmed relation among modular graph functions at the verifier’s own random points (32 digits). Dashed: acceptance threshold 10^{-24} ; dotted: threshold 10^{-14} for relations containing a Laplacian (orange).**Fig. 3** Exact comparison, for every level $N \leq 30$, of $\dim M_4(\Gamma_0(N))$ (grey) with the rank of the span of all holomorphic eta quotients of weight 4 with trivial character (blue).

Statement	Evidence (claim ids)	Level
Section: Introduction	C-fakt-beweis, C-fakt-haertung, C-fakt-konstanten, C-modular-R12, C-modular-R14, C-modular-R17a, C-modular-R6b, C-modular-R6c, C-modular-R6d, C-neg3	computed_rigorous, observed
Section: Setting	C-fakt-beweis, C-modell, C-modular-R14, C-modular-R17a	computed_rigorous
Theorem 1 (Closed forms of the constants, $3 \leq w \leq 61$)	C-fakt-konstanten, C-modular-R17a	computed_rigorous
Section: Exact identities for odd weight	C-fakt-beweis, C-fakt-konstanten, C-modular-R16, C-modular-R17a	computed_rigorous
Proposition 1 (Constants for $27 \leq w \leq 35$)	C-modular-R16	computed_rigorous
Remark 1 (Status of the computer-assisted parts)	C-fakt-beweis, C-modell, C-modular-R9a, C-verfuegbarkeit	computed_rigorous, observed
Theorem 2 (Uniqueness of the harmonic combination, $3 \leq w \leq 25$)	C-modular-R14	computed_rigorous
Section: Uniqueness and Laplace algebra	C-modular-R14, C-modular-R17b, C-modular-R9b	computed_rigorous
Proposition 2 (Laplace equations at $w = 9$ and $w = 11$)	C-modular-R9a, C-modular-R9b	computed_rigorous
Remark 2 (Conjecture 1: beyond the examined range)	C-fakt-konstanten	computed_rigorous
Section: Independent numerical confirmation	C-fakt-haertung, C-fakt-konstanten, C-fakt-kriterium1, C-modell, C-modular-R11a, C-modular-R11b, C-modular-R12, C-modular-R13, C-modular-R17c, C-modular-R17d, C-modular-R6a, C-modular-R6b, C-modular-R6c, C-modular-R6d, C-modular-R7a	computed_rigorous, observed
Example 1 (Weights 3 and 5)	C-modular-R6b, C-modular-R6c	observed
Observation 1 (Weight 7)	C-modular-R6d	observed
Observation 2 (Weight 9)	C-modular-R6a	observed
Observation 3 (Weights 11, 13, 15, 17)	C-modular-R11a, C-modular-R11b, C-modular-R13, C-modular-R7a	observed
Observation 4 (Full weight-11 basis)	C-modular-R12	observed
Observation 5 (Criterion 1 failed, reported openly)	C-fakt-kriterium1	observed
Observation 6 (Hardening of the weight-11 identity (numerical))	C-fakt-haertung	observed
Section: Discussion	C-fakt-beweis, C-fakt-haertung, C-fakt-konstanten, C-fakt-kriterium1, C-modell, C-modular-R12, C-modular-R14, C-modular-R17a, C-neg1, C-neg2, C-neg3, C-neg4	computed_rigorous, observed
Section: Declarations	C-verfuegbarkeit	observed
Section: Appendix A: Numerical methods and error control	C-fakt-beweis, C-methode, C-modell, C-modular-R1, C-modular-R11a, C-modular-R11b, C-modular-R13, C-modular-R17a, C-modular-R1b, C-modular-R1c, C-modular-R1d, C-modular-R4a, C-modular-R4b, C-modular-R4c, C-modular-R4d, C-modular-R5a, C-modular-R5b, C-modular-R5c, C-modular-R5d, C-modular-R5e, C-modular-R5f, C-modular-R5g, C-modular-R5h, C-modular-R8, C-modular-R9a, C-modular-R9b, C-neg1, C-neg2, C-neg3, C-neg4, C-neuheit-suche, C-redteam	computed_rigorous, observed